



رَبِّ اشْرَحْ لِي سَدْرِي وَيَسِّرْ لِي أَمْرِي وَاحْلُلْ عُقْدَةً مِّنْ لِّسَانِي يَفْقَهُوا قَوْلِي

**Rabbish Rahli Sadri wa Yassirli Amri Wahlul Uqadatam
Millisani Yafkahu Qawli.**

My Lord, Uplift my heart for me, make my task easy, and remove the dullness from my tongue so that they may understand me.

(Surah Ta Ha: 25-26-27-28)



Drug Distribution

Pharmacokinetic



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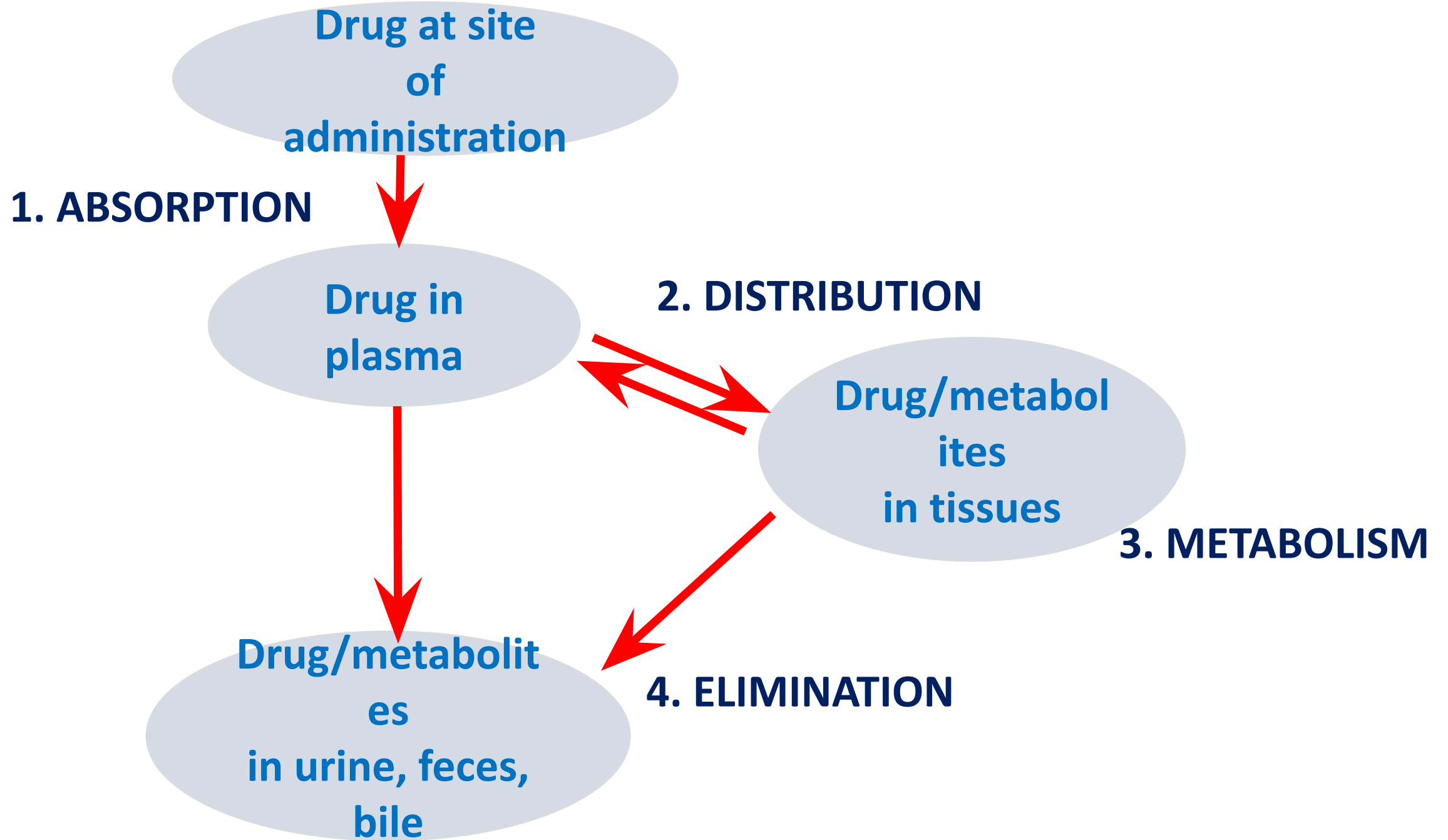
OBJECTIVES

By the end of the lecture the students of BDS 1st year will be able to:

- Define distribution.
- Discuss the factors affecting drug distribution.
- Define volume of distribution.
- Discuss the apparent volume of distribution.

DISTRIBUTION

The process by which the drug reversibly
leaves the blood and enters the tissues



FACTORS MODIFYING DRUG DISTRIBUTION

- **Size of organ**
- **Rate of blood flow**
- **Capillary permeability**
- **Lipophilicity**
- **Binding of drug to proteins**

A. Determinants of Distribution

1. Size of the organ

Larger the size of the organ, larger will be the distribution.

2. Blood flow

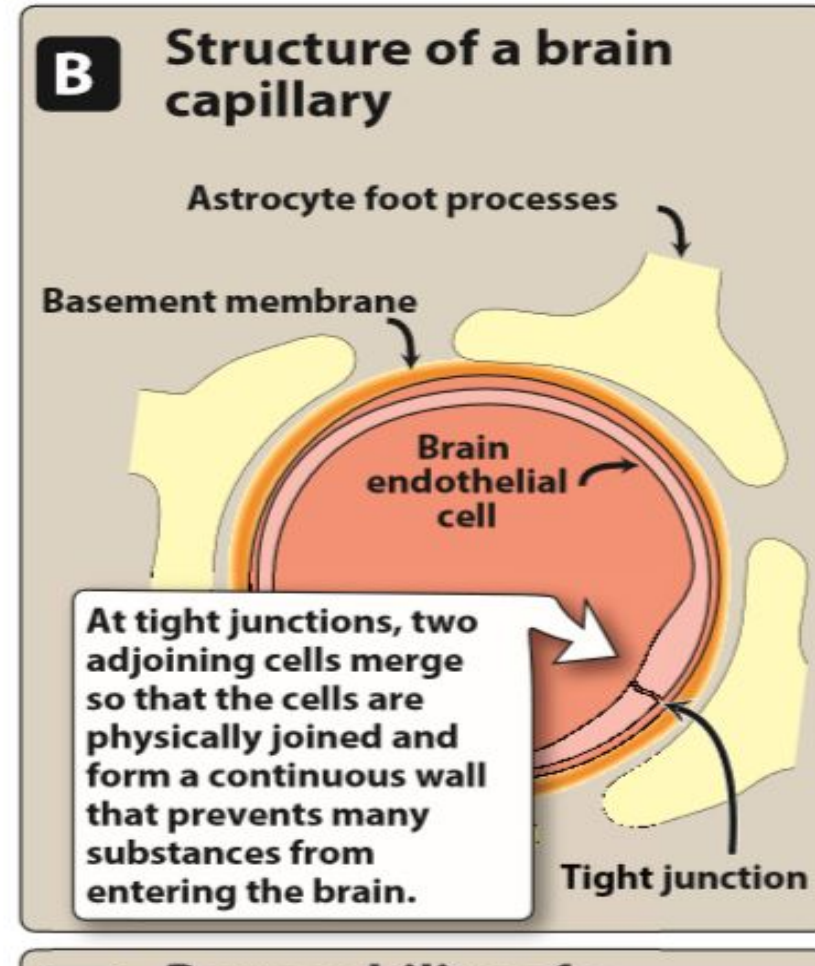
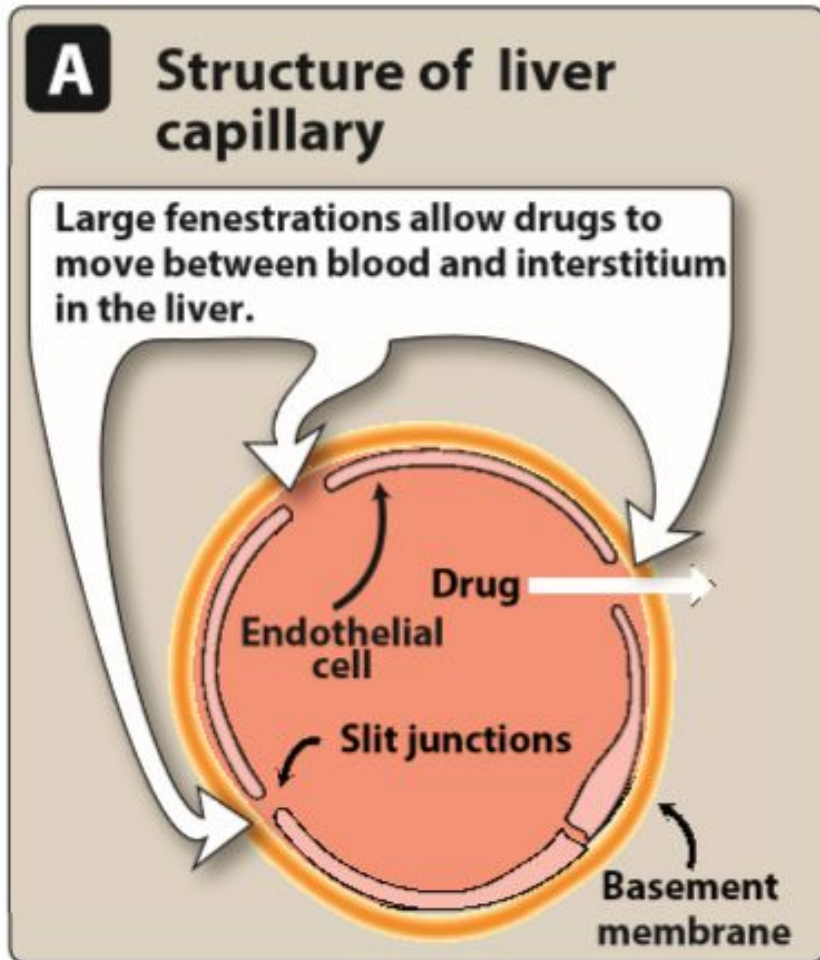
Well-perfused tissues usually achieve high tissue concentrations sooner than poorly perfused tissues.

A. Determinants of Distribution

3. Capillary permeability

- Capillary structure varies in terms of the fraction of the basement membrane exposed by slit junctions between endothelial cells.
- In the liver and spleen, a significant portion of the basement membrane is exposed due to large, discontinuous capillaries through which large plasma proteins can pass.
- In the brain, the capillary structure is continuous, and there are no slit junctions.
- To enter the brain, drugs must pass through the endothelial cells of the CNS capillaries or be actively transported.

Capillary permeability



A. Determinants of Distribution

4. Solubility

For example, some organs (such as the brain) have a high lipid content and thus dissolve a high concentration of lipid-soluble agents rapidly.

5. Lipophilicity

- The chemical nature of a drug strongly influences its ability to cross cell membranes.
- Lipophilic drugs readily move across most biologic membranes. These drugs dissolve in the lipid membranes and penetrate in the cell .

A. Determinants of Distribution

6. Binding

Binding of a drug to macromolecules in the blood or a tissue compartment tends to increase the drug's concentration in that compartment.

For example, warfarin is strongly bound to plasma albumin, which restricts warfarin's diffusion out of the vascular compartment.

Conversely, chloroquine is strongly bound to extravascular tissue proteins, which results in a marked reduction in the plasma concentration of chloroquine.

Binding of drugs to plasma proteins and tissues

1. Binding to plasma proteins:

Reversible binding to plasma proteins sequesters drugs in a nondiffusible form and slows their transfer out of the vascular compartment.

- **Albumin** is the major drug-binding protein and may act as a **drug reservoir** (as the concentration of free drug decreases due to elimination, the bound drug dissociates from the protein).
- This maintains the free drug concentration as a constant fraction of the total drug in the plasma.

Binding of drugs to plasma proteins and tissues

2. Binding to tissue proteins:

Many drugs accumulate in tissues, leading to higher concentrations in tissues than in the extracellular fluid and blood.

Drugs may accumulate as a result of binding to lipids, proteins, or nucleic acids.

Drugs may also be actively transported into tissues.

- **Tissue reservoirs** may serve as a major source of the drug and prolong its actions or cause local drug toxicity. (For example, acrolein, the metabolite of cyclophosphamide, can cause hemorrhagic cystitis because it accumulates in the bladder.)

VOLUME OF DISTRIBUTION

The volume of distribution (Vd) relates the amount of drug in the body to the plasma concentration according to the following equation:

$$V = \frac{\text{Amount of drug in the body}}{\text{Plasma drug concentration}}$$

(Units Volume)

VOLUME OF DISTRIBUTION

A simple approximate rule for the aqueous compartments of the normal body is as follows:

40% of total body weight is intracellular water and

20% is extracellular water;

thus, water constitutes approximately 60% of body weight.

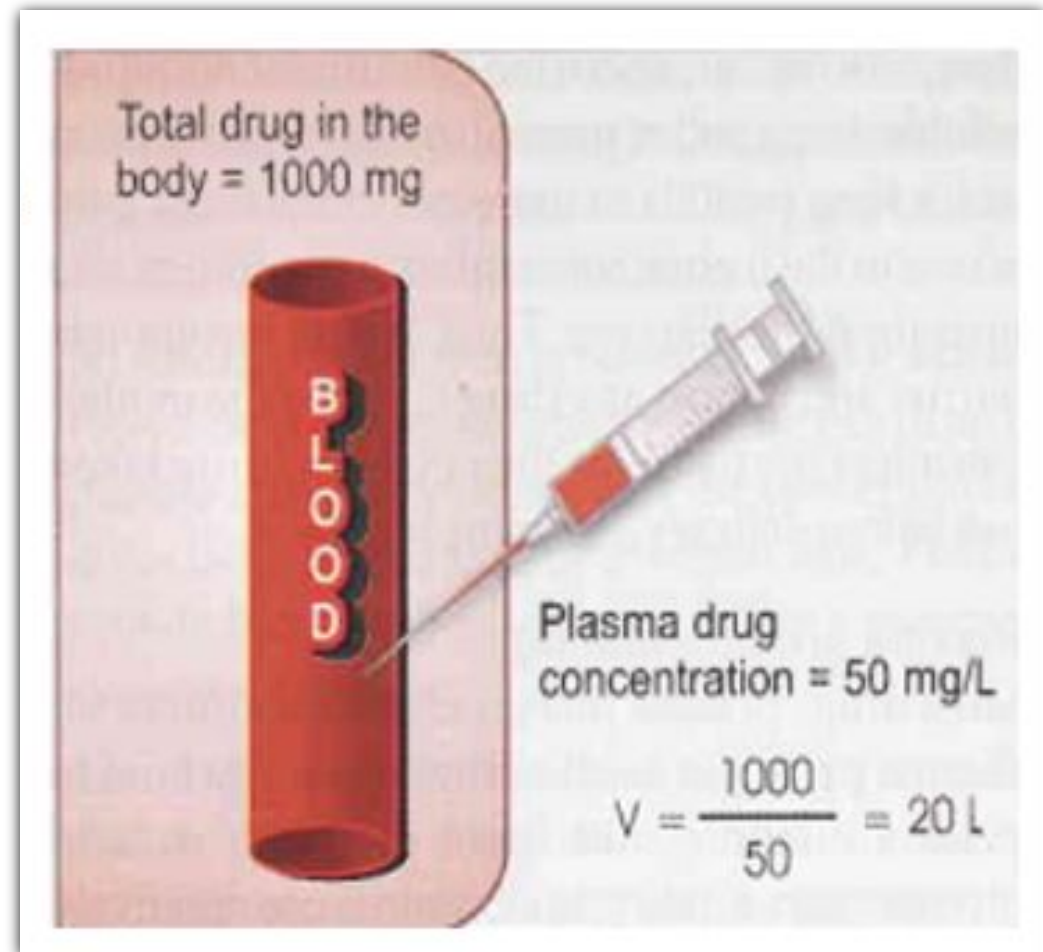
VOLUME OF DISTRIBUTION

Total Body Water = 42L

Intracellular = 28L

Extracellular = 14L

**(10L Interstitial Fluid + 4 L
Plasma Volume)**



B. Apparent Volume of Distribution and Physical Volumes

Distribution into the water compartments in the body:

Once a drug enters the body, it has the potential to distribute into any one of the three functionally distinct compartments of body water or to become sequestered in a cellular site.

a. Plasma compartment:

If a drug has a high molecular weight or is extensively protein bound, it is too large to pass through the slit junctions of the capillaries and, thus, is effectively trapped within the plasma compartment. As a result, it has a low V_d that approximates the plasma volume or about 4 L in a 70-kg individual.

FOR EXAMPLE Heparin

B. Apparent Volume of Distribution and Physical Volumes

b. Extracellular fluid:

If a drug has a low molecular weight but is hydrophilic, it can pass through the endothelial slit junctions of the capillaries into the interstitial fluid.

However, hydrophilic drugs cannot move across the lipid membranes of cells to enter the intracellular fluid.

Therefore, these drugs distribute into a volume that is the sum of the plasma volume and the interstitial fluid, which together constitute the extracellular fluid (about 20% of body weight or 14 L in a 70-kg individual).

FOR EXAMPLE Aminoglycoside antibiotics.

B. Apparent Volume of Distribution and Physical Volumes

c. Total body water:

If a drug has a low molecular weight and is lipophilic, it can move into the interstitium through the slit junctions and also pass through the cell membranes into the intracellular fluid.

These drugs distribute into a volume of about 60% of body weight or about 42 L in a 70-kg individual.

FOR EXAMPLE Ethanol

QUIZ

- A drug is administered at a dose of 500 mg i/v. Immediately afterwards, the plasma level of the drug comes out to be 0.01 mg /ml.

What is the apparent volume of distribution?

Vd= (apparent) amount of drug in body/ plasma C
= 500/0.01= 50,000 ml.

THANKS